

Public Cloud Services Migration Workflow



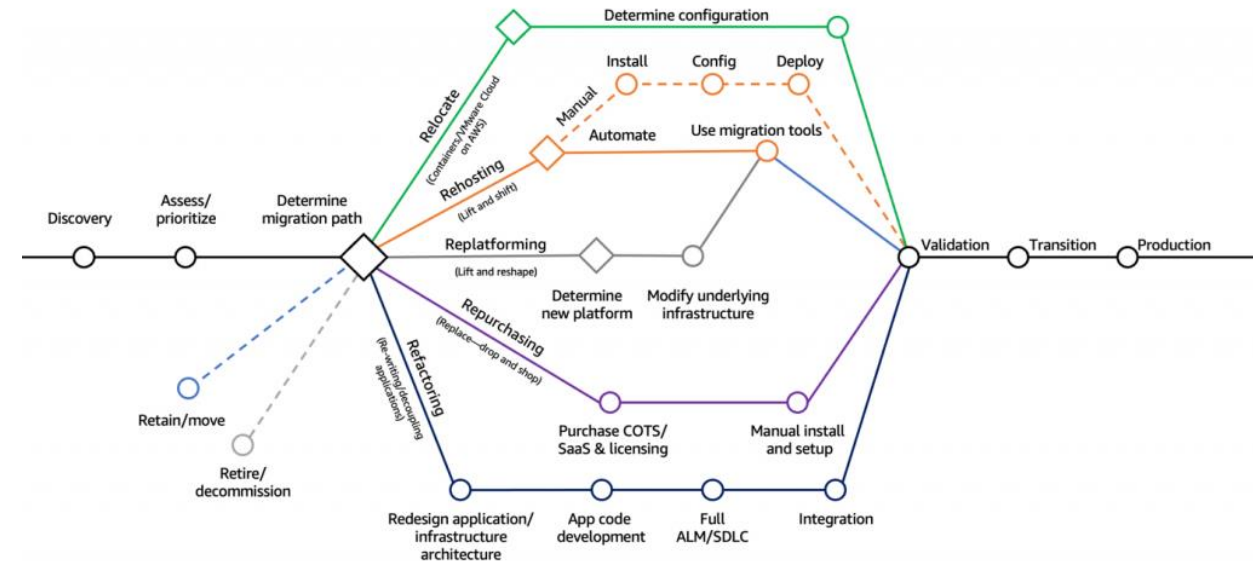
Cloud Migration going wrong



High Level Strategy: The 5 R's of Gartner and the 7 R's of AWS

Originated by Gartner:

- Rehost (lift and shift)
- Refactor (optimize with cloud services)
- Rearchitect (redesign for cloud-native features)
- Rebuild (create new applications for the cloud)
- Replace (adopt SaaS solutions).



Technical Basics of Cloud Migration

Given a containerized environment...what has to be done?

1. Create target platform for containerized Environment
2. Migration plan
3. Perform the migration
 1. Change of access
 2. Continuous data sync

1. Target Platform

Given a containerized environment:

- Targetplatform should host containers only on AWS: → AWS/ECS

Steps to be taken:

1. Ensure container can run on AWS/ECS manually
2. Create AWS/ECS platform with IaC of your choice. You might use any tooling you want, however you need to explain your choice and the setup

Be aware of the following points:

1. You might need an additional configuration-mgmt layer.
2. You need to generate a Token in gitlab.fhnw.ch for access your images. The token should be constraint on your project and be focused on the registry only. Make a short research how to handle it securely.

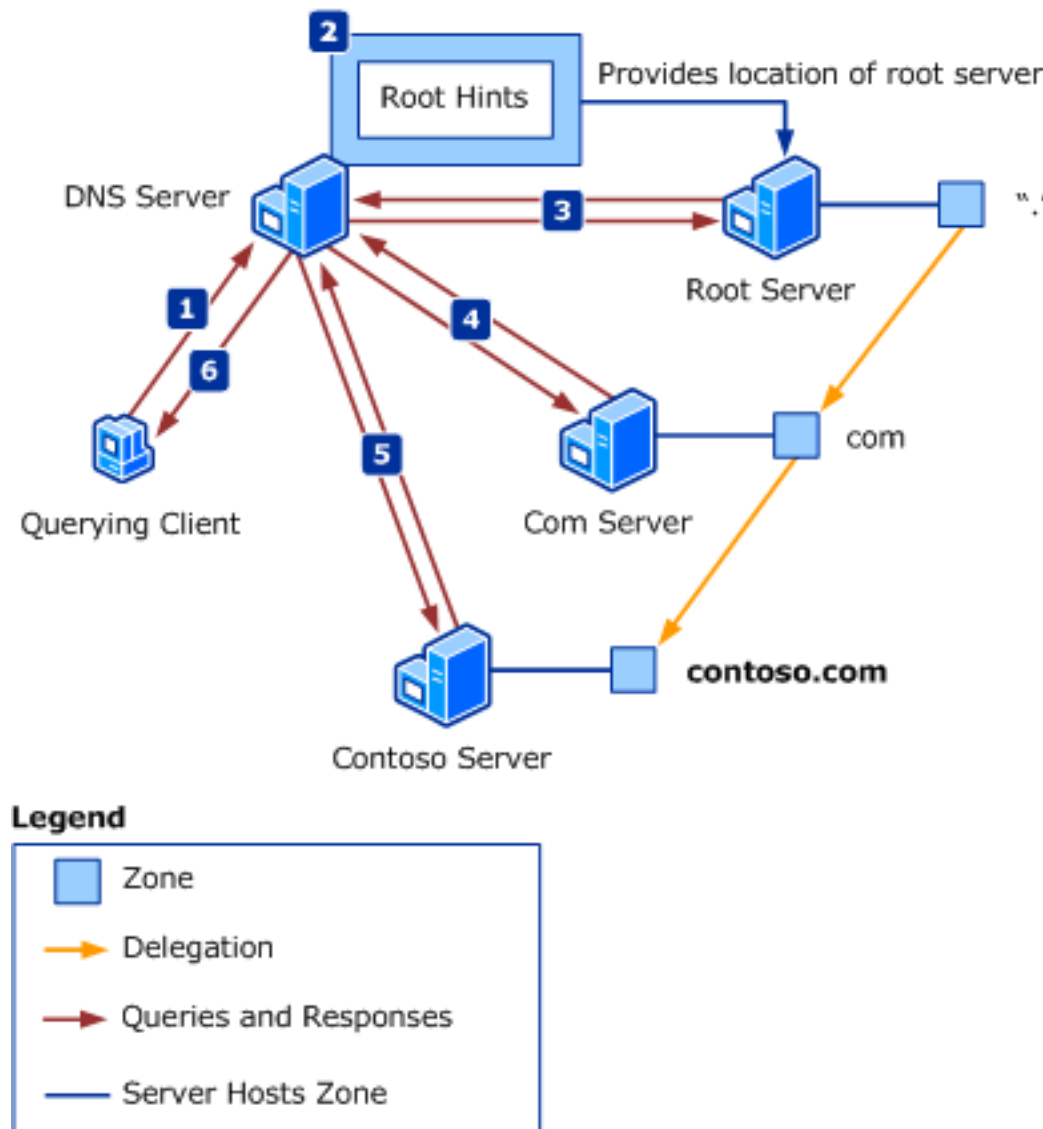
2. Migration Plan

Given an automatic provisioned and containerized environment on AWS/ECS:

Steps to be taken:

1. Make yourself familiar with DNS
 1. Reduce the TTL for migration.
 2. Define an endpoint for your AWS/ECS endpoint (also in IaC)
2. Sketch the migration by switching the endpoint from Switch/IaaS to AWS/ECS. Write down the concrete steps as runbook including Snippets and checks.

DNS-Basics



<https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/plan/reviewing-dns-concepts>

3. Performing the migration

Given an automatic provisioned and containerized environment on AWS/ECS and a migration plan

Steps to be taken:

1. Define a point in time for migration
2. Record your screencast of the migration and show in the screencast how seamless the application is migrated
3. Bonustask

Reflect the following question: What additional steps would be necessary if a state would have been included in the transfer?

